

Tide retrospective: the local Taylor comparison (smooth-germ programme, stage C3)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The stage the scoping consult designated the programme’s hardest: the normalized local Taylor comparison, where the smooth-versus-polynomial analysis actually lives. A dedicated shape consult settled three design choices before any Lean: the moving radius $r(q) = \sqrt{q}$ in place of $t^{-2/5}$ (outer factor $e^{-c/q}$, no `rpow` anywhere in the proof), the global side condition $2 \leq D$ defusing natural-number subtraction in every exponent, and a direct `Metric.tendsto_nhds` epsilon-of-room ending whose quantifier order — jet- ε chosen from the target η first, the q -window shrunk second — matches the statement being proved.

2 The thing formalised

Six declarations in `Laplace/OneD/TaylorCompare.lean`. The admissibility package (continuity, normalized minimum, global lower and local upper quadratic envelopes) with its order bounds: the partition lower bound $Z \geq 2\delta e^{-\kappa\delta^2} q$ — restrict to $|x| \leq q\delta$, where the scaling relation makes the exponent at most $\kappa\delta^2$ — and the signed moment bound $|A_s| \leq C_s q^{s+1}$. Then the two comparison theorems: for admissible K_1, K_2 with $K_1 - K_2 = o(|x|^D)$ at 0 in epsilon-radius form,

$$A_s(K_1, q) - A_s(K_2, q) = o(q^{s+D-1}), \quad F_s(K_1, q) - F_s(K_2, q) = o(q^{s+D-2}),$$

along $q \rightarrow 0^+$. The unnormalized proof splits at $|x| = \sqrt{q}$: inside, the exponential secant bound and the jet hypothesis give the majorant $\varepsilon_j t |x|^{s+D} e^{-t\rho x^2}$ — the Taylor power kept *inside* the Gaussian integral, per the scoping consult’s warning that sup-ing it loses powers as R grows — integrating to $\varepsilon_j C q^{s+D-1}$; outside, two envelope tail bounds with $e^{-\rho\nu/(2q)}$ prefactors, each vanishing against any power.

3 The quotient step is not the generic lemma

The weighted-jet programme’s quotient-difference lemma requires nonvanishing denominator limits, which holds in rescaled u -space but fails here: in x -space both partition functions tend to 0 like q . The normalized comparison instead uses the bespoke decomposition $F_1 - F_2 = (A_1 - A_2)/Z_1 + A_2(Z_2 - Z_1)/(Z_1 Z_2)$ squeezed against the two unnormalized limits via $Z \geq C_0 q$ and $|A| \leq C q^{s+1}$ — an instructive case of a reusable lemma almost, but not quite, fitting.

4 Roadblocks and resolutions

The substantive proofs went through close to the consult’s script; the friction was tactical and two items are new catalogue entries. A new instance of the cascading-rewrite class: `show s + D = (s+D-2)+2` rewrote the `s + D` *inside* `s + D - 2`, leaving unprovable residues — `nth_rewrite` on a named equation is the fix, the exponent-arithmetic analogue of the radical-atom rule. And `gcongr` discharges monotone-denominator side goals from ambient hypotheses entirely, so trailing `exacts` die with “no goals” even when they carry the substantive inequality. Familiar classes: `positivity` cannot consume context nonnegativity of `set` constants; beta-unreduced congruence goals; dead tactics after closing `field.simp`; the heartbeat budget on the assembled calc.

5 Deliberation and check

The shape consult is archived in the tide log with the votes. The statement-level numerical check ran before formalisation: for $K_1 = x^2 + 0.3x^5e^{-x^2}$, $K_2 = x^2$, $D = 4$, the scaled difference $(A_2(K_1) - A_2(K_2))/q^5$ decayed $0.0126 \rightarrow 0.0011 \rightarrow 0.00008$ over $q = 0.2, 0.1, 0.05$ — $o(q^{s+D-1})$ with margin (the true next order is one further power of q).

6 What the next tide inherits

Stage C4: the Taylor/stabilizer adapter — construct the degree- D Taylor polynomial of a smooth admissible loss, verify the Peano remainder supplies exactly this tide’s epsilon-radius jet hypothesis against the stabilized polynomial, and expose the C1-compatible scaled data; then C5 assembles the finite-order smooth recovery. The comparison machinery here is stated for arbitrary admissible potentials, so C4 is composition plus one genuinely new ingredient: the Peano remainder from `iteratedDeriv` data.